

Programme	B. Sc. Mathematics Honours			
Course Code	MAT1MN106			
Course Title	PRINCIPLES OF MICRO ECONOMICS			
Type of Course	Minor			
Semester	I			
Academic Level	100 - 199			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Higher Secondary Mathematics			
Course Summary	Explore market behaviour in Demand and Supply Analysis, focusing on utility, the law of demand, supply, and elasticity, and delve into Cost and Revenue Functions to understand cost structures, revenue functions, and their relation to demand elasticity. Explore the Theory of Consumer Behaviour to comprehend utility maximization and rational consumer choices, then apply economic optimization techniques using derivatives in Economic Applications to optimize functions and solve constrained optimization problems efficiently.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse the factors affecting demand and supply and determine market equilibrium.	An	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO2	Apply the concepts of cost and revenue functions to analyze short-run and long-run production decisions.	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO3	Evaluate economic functions and optimize using derivatives and Lagrange multipliers.	E	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		1. Principles of Micro Economics, H.L.Ahuja, 15 th revised edition, S.Chand 2. Introduction to Mathematical Economics, Edward.T.Dowling, 3 rd edition, Schaum's Outline series, TMH		
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)
I	Demand and Supply Analysis Text(1) (Relevant sections of chapter 5 and 7)		13	Min 15
	1	Utility and demand, the meaning of demand and quantity demanded		
	2	The law of demand- demand curve- market demand curve		
	3	Reasons for the law of demand- slope of a demand curve		
	4	Shift in demand- demand function and demand curve		
	5	The meaning of supply- supply function- law of supply		
	6	Slope of a supply curve- shift in supply- market equilibrium		
	7	Price elasticity of demand- measurement of price elasticity- arc elasticity of demand- cross elasticity of demand		
II	Cost and Revenue Functions Text (1) (Relevant sections of chapter 19 and 21)		12	Min 15
	8	Cost function- Average Cost (AC) and Marginal Cost (MC)		
	9	Short run costs: Total Fixed and Variable Cost - Short Run average cost curve- Average Variable Cost (AVC)- Relationship between AVC and Average product- Average Total Cost- Marginal Cost		
	10	Long run costs: Long Run Average Cost Curve- relationship of Long run Average Cost Curve (LAC) and Long run Marginal Cost Curve (LMC) with SAC and SMC		
	11	Revenue function, Marginal Revenue (MR) and Average Revenue (AR)		
	12	Relation between MR, AR and elasticity of demand		
III	Theory Of Consumer Behaviour Text (1) (Relevant sections of chapter 9 and 11)		10	Min 15
	13	Cardinal utility analysis- the law of diminishing marginal utility- illustration of law of diminishing marginal utility		
	14	The law of equi-marginal utility		
	15	Indifference curves- ordinal utility		
	16	Marginal rate of substitution- properties of indifference curves		
	IV	Economic Applications of Derivatives Text (2) (Chap-4: sec 4.7&4.8, Chap 5: sec 5.1 to 5.7)		
17		Economic application of derivatives- marginal, average, total concepts		

	18	Optimizing economic function		Min 15
	19	Functions of several variables and partial derivatives		
	20	Second order partial derivatives, optimization of multivariable function		
	21	Constrained optimization with Lagrange multipliers		
	22	Significance of Lagrange multipliers, differentials		
V	Open Ended		12	
	Derivative of a function, first order derivative, second order derivative, local maxima, local minima, optimization			
References:				
1. Mathematical analysis for economists, RGD Allen, Macmillan.				
2. Maths for Economics(3/e), Geoff Renshaw, Oxford University Press, N.Y. (2012)				

Note: 1) Proofs of all the results are exempted for external exam. (2) 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module. Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	2	3	2	3	1	2
CO 2	2	1	3	2	3	2	3	1	2
CO 3	3	2	3	1	3	2	3	1	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓